BirthdaySim:

Input: 23 people in the classroom, 10,000 trials

```java
BirthdaySim b = new BirthdaySim();
//number of people in the class, number of trials
b.doBirthdaySimulation(23, 10000);
```

Output:

```
Problems  @ Javadoc  Declaration  Console ×
<terminated> MonteCarloTester [Java Application] C:\Users\sophi\.p2\pool\plugins\org.eclipse.justj.openjdk.h
Probability of at least two people sharing a birthday: 50.88%
```

Comparison to probability found using math:

P(no shared birthdays) = $\frac{365 \times 364 \times \ldots \times 343}{(365)^{23}}$ = .492703

P(shared birthday) = 1 – P(no shared birthdays) = .507297 = 50.73%

Input: 15 people in the classroom, 10,000 trials

```java
BirthdaySim b = new BirthdaySim();
//number of people in the class, number of trials
b.doBirthdaySimulation(15, 10000);
```

Output:

```
Problems  @ Javadoc  Declaration  Console ×
<terminated> MonteCarloTester [Java Application] C:\Users\sophi\.p2\pool\plugins\org.e
Probability of at least two people sharing a birthday: 25.14%
```

Comparison to probability found using math:

P(no shared birthdays) = $\frac{365 \times 364 \times \ldots \times 351}{(365)^{15}}$ = .747099

P(shared birthday) = 1 – P(no shared birthdays) = .252901 = 25.29%

Input: 32 people in the classroom, 10,000 trials

```java
BirthdaySim b = new BirthdaySim();
//number of people in the class, number of trials
b.doBirthdaySimulation(32, 10000);
```

Output:

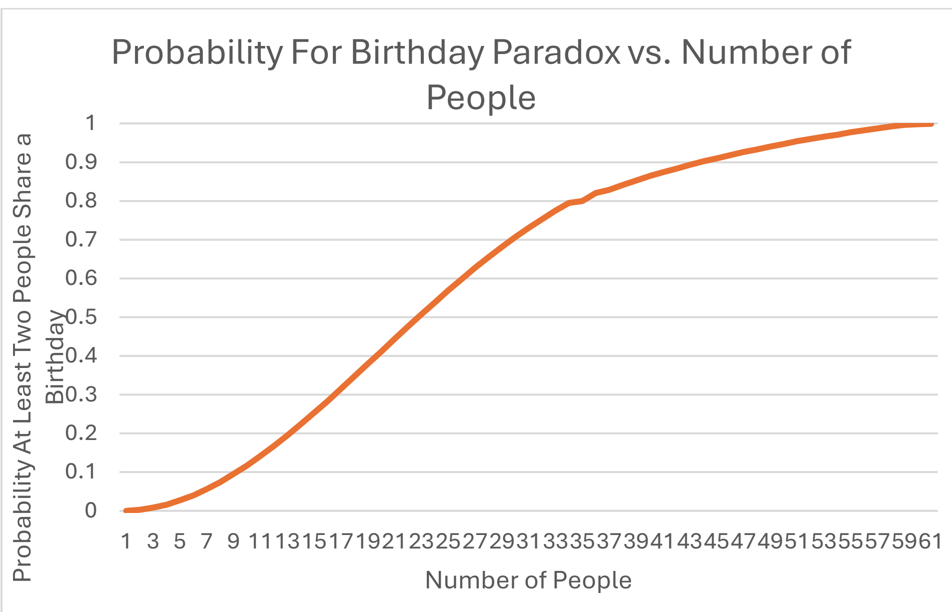
```
Problems  @ Javadoc  Declaration  Console ×
<terminated> MonteCarloTester [Java Application] C:\Users\sophi\.p2\pool\plugins\org.eclipse.justj.op
Probability of at least two people sharing a birthday: 75.27000000000001%
```

Comparison to probability found using math:

P(no shared birthdays) = $\frac{365 \times 364 \times . . . \times 334}{(365)^{32}}$ = .246652

P(shared birthday) = 1 – P(no shared birthdays) = .753348 = 75.33%

Graph showing the probability distribution for Birthday Paradox



From around the 1st person to the 25th person, the probability of having a shared birthday is increasing at an increasing rate. After that, we switch concavity, and the probability starts increasing at a decreasing rate as the curve flattens out near 100% probability.

Mulligan:

Input: 10,000 trials

```java
        Mulligan m = new Mulligan();
        //number of trials
        m.doMulliganSim(10000);
```

Output:

```
Problems  Javadoc  Declaration  Console
<terminated> MonteCarloTester [Java Application] C:\Users\sophi\.p2\pool\plugins\org.eclipse.justj.openjdk.hots
Probability of drawing a mulligan with 1 Pokemon cards is 88.26%
Probability of drawing a mulligan with 2 Pokemon cards is 78.38000000000001%
Probability of drawing a mulligan with 3 Pokemon cards is 68.83%
Probability of drawing a mulligan with 4 Pokemon cards is 59.730000000000004%
Probability of drawing a mulligan with 5 Pokemon cards is 51.35999999999999%
Probability of drawing a mulligan with 6 Pokemon cards is 46.53%
Probability of drawing a mulligan with 7 Pokemon cards is 39.48%
Probability of drawing a mulligan with 8 Pokemon cards is 34.949999999999996%
Probability of drawing a mulligan with 9 Pokemon cards is 30.159999999999997%
Probability of drawing a mulligan with 10 Pokemon cards is 25.88%
Probability of drawing a mulligan with 11 Pokemon cards is 22.15%
Probability of drawing a mulligan with 12 Pokemon cards is 19.25%
Probability of drawing a mulligan with 13 Pokemon cards is 16.689999999999998%
Probability of drawing a mulligan with 14 Pokemon cards is 14.180000000000001%
Probability of drawing a mulligan with 15 Pokemon cards is 11.49%
Probability of drawing a mulligan with 16 Pokemon cards is 10.09%
Probability of drawing a mulligan with 17 Pokemon cards is 8.15%
Probability of drawing a mulligan with 18 Pokemon cards is 6.950000000000001%
Probability of drawing a mulligan with 19 Pokemon cards is 5.86%
Probability of drawing a mulligan with 20 Pokemon cards is 4.8500000000000005%
Probability of drawing a mulligan with 21 Pokemon cards is 4.21%
Probability of drawing a mulligan with 22 Pokemon cards is 3.16%
Probability of drawing a mulligan with 23 Pokemon cards is 2.74%
Probability of drawing a mulligan with 24 Pokemon cards is 2.06%
Probability of drawing a mulligan with 25 Pokemon cards is 1.7500000000000002%
Probability of drawing a mulligan with 26 Pokemon cards is 1.47%
Probability of drawing a mulligan with 27 Pokemon cards is 1.0999999999999999%
Probability of drawing a mulligan with 28 Pokemon cards is 0.8699999999999999%
Probability of drawing a mulligan with 29 Pokemon cards is 0.62%
Probability of drawing a mulligan with 30 Pokemon cards is 0.5%
Probability of drawing a mulligan with 31 Pokemon cards is 0.41000000000000003%
Probability of drawing a mulligan with 32 Pokemon cards is 0.32%
Probability of drawing a mulligan with 33 Pokemon cards is 0.25%
Probability of drawing a mulligan with 34 Pokemon cards is 0.16999999999999998%
Probability of drawing a mulligan with 35 Pokemon cards is 0.13999999999999999%
Probability of drawing a mulligan with 36 Pokemon cards is 0.1%
Probability of drawing a mulligan with 37 Pokemon cards is 0.09%
Probability of drawing a mulligan with 38 Pokemon cards is 0.03%
Probability of drawing a mulligan with 39 Pokemon cards is 0.04%
Probability of drawing a mulligan with 40 Pokemon cards is 0.0%
Probability of drawing a mulligan with 41 Pokemon cards is 0.04%
Probability of drawing a mulligan with 42 Pokemon cards is 0.01%
```

```
Probability of drawing a mulligan with 43 Pokemon cards is 0.01%
Probability of drawing a mulligan with 44 Pokemon cards is 0.0%
Probability of drawing a mulligan with 45 Pokemon cards is 0.0%
Probability of drawing a mulligan with 46 Pokemon cards is 0.0%
Probability of drawing a mulligan with 47 Pokemon cards is 0.0%
Probability of drawing a mulligan with 48 Pokemon cards is 0.0%
Probability of drawing a mulligan with 49 Pokemon cards is 0.0%
Probability of drawing a mulligan with 50 Pokemon cards is 0.0%
Probability of drawing a mulligan with 51 Pokemon cards is 0.0%
Probability of drawing a mulligan with 52 Pokemon cards is 0.0%
Probability of drawing a mulligan with 53 Pokemon cards is 0.0%
Probability of drawing a mulligan with 54 Pokemon cards is 0.0%
Probability of drawing a mulligan with 55 Pokemon cards is 0.0%
Probability of drawing a mulligan with 56 Pokemon cards is 0.0%
Probability of drawing a mulligan with 57 Pokemon cards is 0.0%
Probability of drawing a mulligan with 58 Pokemon cards is 0.0%
Probability of drawing a mulligan with 59 Pokemon cards is 0.0%
Probability of drawing a mulligan with 60 Pokemon cards is 0.0%
```

We will select 3 iterations and check the results of the simulation against results obtained using combinations

1. 1 Pokemon and 59 Energy:

   N = $C_7^{60} = \frac{60!}{7!(60-7)!} = 386206920$

   $n_a = C_7^{59} = \frac{59!}{7!(59-7)!} = 341149446$

   $\frac{n_a}{N} = .8833333 = 88.33\%$

   This compares very closely to the Monte Carlo value, percent error = .079%

2. 20 Pokemon and 40 Energy:

   N = $C_7^{60} = \frac{60!}{7!(60-7)!} = 386206920$

   $n_a = C_7^{40} = \frac{40!}{7!(40-7)!} = 18643560$

   $\frac{n_a}{N} = .0482735 = 4.83\%$

   This compares very closely to the Monte Carlo value, percent error = .414%

3. 40 Pokemon and 20 Energy:

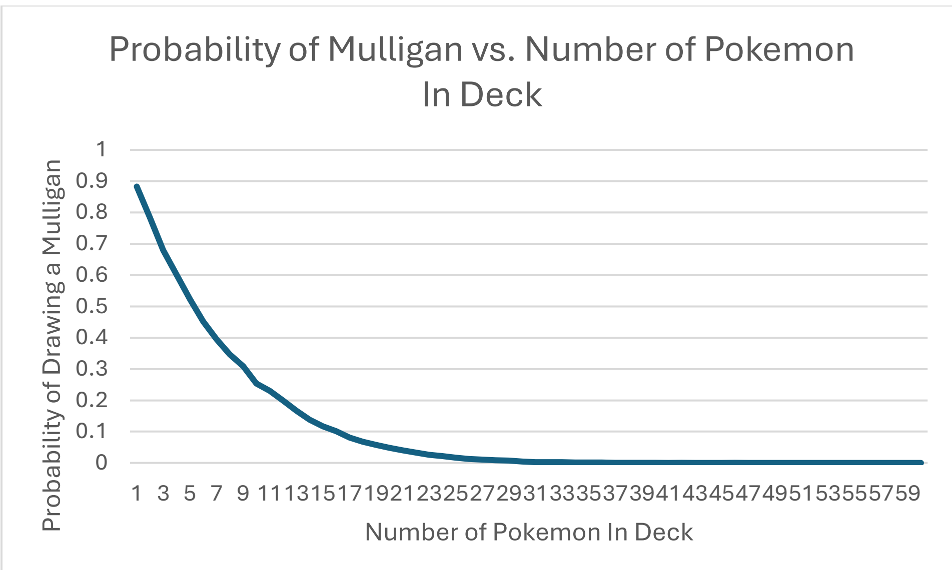
N = $C_7^{60} = \frac{60!}{7!(60-7)!} = 386206920$

$n_a = C_7^{20} = \frac{20!}{7!(20-7)!} = 77520$

$\frac{n_a}{N}$ = .000200721 = .020%

This compares very closely to the Monte Carlo value, percent error = .032%

Graph showing probability distribution for drawing a mulligan:



This graph can give us important information as a player. The probability of drawing a mulligan decreases very quickly as we add more Pokemon to the deck. So here, we can see that a player really only needs about 20 Pokemon to ensure that they most likely will never draw a Mulligan.

ThreeDoors:

Input: 10,000 trials

```java
        ThreeDoors d = new ThreeDoors();
        //number of trials
        d.doDoorSimulation(10000);

```

Output:

```
Problems  Javadoc  Declaration  Console ×
<terminated> MonteCarloTester [Java Application] C:\Users\sophi\.p2\poc
If contestant stays, chance of winning is 33.25%
If contestant switches, chance of winning is 66.29%
```

In the Monty Hall problem, players are shown to double their odds of winning from 33.33% to 66.66% if they choose to switch after the first door is revealed. We will use a table to show why this is true.

Assume the initial selection the player chose is Door 1

Door 1	Door 2	Door 3	Staying
Goat	Goat	Car	Lose
Goat	Car	Goat	Lose
Car	Goat	Goat	Win

Here, we can see that the player will win 1/3 times = 33.33%

Assume the initial selection the player chose is Door 1, and then they switch

Door 1	Door 2	Door 3	Switching
Goat	Goat	Car	Win
Goat	Car	Goat	Win
Car	Goat	Goat	Lose

Here, we can see that the player will win 2/3 times = 66.66%

RareCandySim:

Input: 10,000 trials

```
        RareCandySim r = new RareCandySim();
        //number of trials
        r.doRareCandySim(10000);

```

Output:

```
The probability of all 1 rare candies in prize pile: 10.39%
The probability of all 2 rare candies in prize pile: 0.84%
The probability of all 3 rare candies in prize pile: 0.06%
The probability of all 4 rare candies in prize pile: 0.0%
```

We will show how accurately the Monte Carlo simulation is to the calculated probability

1. 1 Rare Candy:

   P(rare candy was not drawn in the hand) = $\frac{\binom{59}{7}}{\binom{60}{7}} = \frac{53}{60}$

   P(rare candy in prize pile | not in hand) = $\frac{\binom{52}{5}}{\binom{53}{6}} = \frac{6}{53}$

   P(rare candy was drawn in the prize pile) = $\frac{53}{60}\left(\frac{6}{53}\right) = \frac{1}{10}$ = .1

   So, the probability that the rare candy will be in the prize pile is 10%

   Percent error: 3.9%

2. 2 Rare Candies:

   P(rare candies were not drawn in the hand) = $\frac{\binom{58}{7}}{\binom{60}{7}} = \frac{689}{885}$

   P(both rare candies in prize pile | both not in hand) = $\frac{\binom{51}{4}}{\binom{53}{6}} = \frac{15}{1378}$

   P(rare candies were drawn in the prize pile) = $\frac{689}{885}\left(\frac{15}{1378}\right) = \frac{1}{118} = .00847$

   So, the probability that both rare candies will be in the prize pile is .847%

   Percent error: .826%

3. 3 Rare Candies:

P(rare candies were not drawn in the hand) = $\frac{\binom{57}{7}}{\binom{60}{7}} = \frac{11713}{17110}$

P(3 rare candies in prize pile | 3 not in hand) = $\frac{\binom{50}{3}}{\binom{53}{6}} = \frac{10}{11713}$

P(rare candies were drawn in the prize pile) = $\frac{11713}{17110}\left(\frac{10}{11713}\right) = \frac{1}{1711} = .000584$

So, the probability that all 3 rare candies will be in the prize pile is .058%

Percent error: 3.448%

4. 4 Rare Candies:

P(rare candies were not drawn in the hand) = $\frac{\binom{56}{7}}{\binom{60}{7}} = \frac{58565}{97527}$

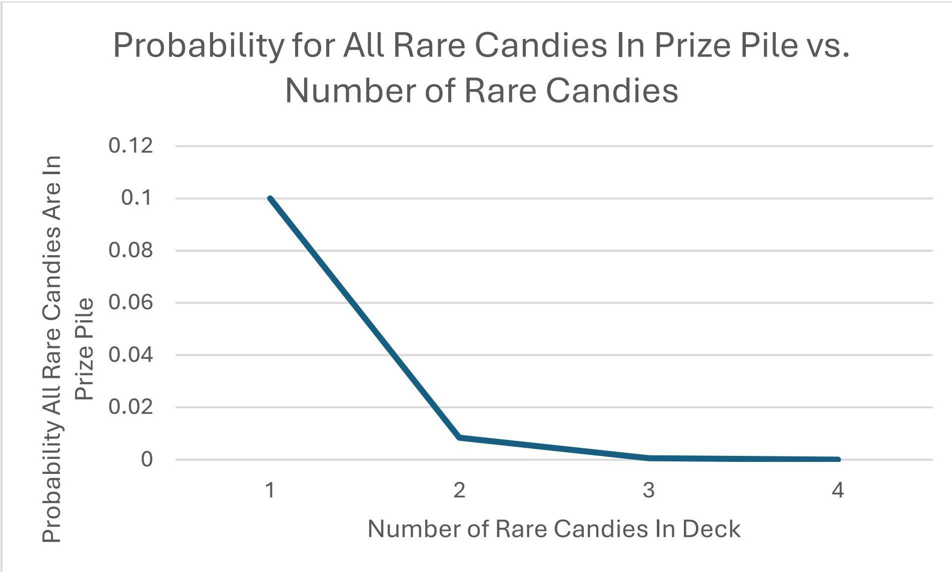
P(4 rare candies in prize pile | 4 not in hand) = $\frac{\binom{49}{2}}{\binom{53}{6}} = \frac{3}{58565}$

P(rare candies were drawn in the prize pile) = $\frac{58565}{97527}\left(\frac{3}{58565}\right) = \frac{1}{32509} = \ .0000308$

So, the probability that all 3 rare candies will be in the prize pile is .00308%

Percent error: 2.600%

Graph showing probability distribution for all rare candies ending up in prize pile:



So, if a player were to rely on their rare candies to evolve their Pokemon, at least 2 rare candies would ensure they most likely will not lose automatically.